CS Rubric - Tech Employment in the Age of AI: Forecasting What's Next

A DS 4002 Case Study by Jensen Harvey

Submission format:

- Upload a link to your GitHub repository to Canvas
- Submit a PDF of your written report alongside the repo link

Group Assignment (recommended teams of 2-3)

General Description: Build a time series forecasting project that meaningfully extends prior work on tech sector layoffs. You'll select one direction for extension, defend your choice, execute the analysis, and report your findings in a polished writeup.

Preparatory Assignments: Time series fundamentals (ARIMA, SARIMA, stationarity, ACF/PACF interpretation). Familiarity with R and the forecast package is helpful but not required.

Why am I doing this? Time series forecasting is one of the most widely used techniques in industry, from finance to public health to workforce planning. This project gives you practice making the kind of judgment calls real analysts face every day: where to source data, which covariates to include, how to validate a model honestly, and how to communicate uncertainty to a non-technical audience. You'll also engage with a topic that directly affects your own career.

- *Course Learning Objective: design and execute a complete time series forecasting workflow on real-world data.*

What am I going to do? Pick one of the extension directions listed below (or propose your own and clear it with your instructor). Build your analysis on top of the starter materials in the repo. Produce a full report documenting your data, methods, findings, and limitations.

Tips for success:

- Start with EDA. The starter materials show you what the previous team found. Confirm it for yourself before extending.
- Be honest about what your model can and can't do. A clear-eyed limitations section is worth more than an overstated result.
- Compare to a baseline. Always. A model that beats nothing isn't really a model.
- Talk to your instructor early about your chosen extension direction.

How will I know I have succeeded? You will meet expectations when you follow the criteria below.

Spec Category	Spec Details
Topic Selection	Pick one of the five extension directions: (1) build a richer dataset by combining Layoffs.fyi with BLS unemployment data and Census records to address public reporting bias; (2) extend to a SARIMAX model with macroeconomic covariates such as Fed funds rate, VC funding volumes, or NASDAQ index; (3) re-evaluate the original model using full 2025 data once available and produce updated RMSE estimates; (4) build industry-level subgroup models for Consumer, Retail, Transportation, or another dominant sector and compare results across verticals; (5) validate the original model's Q1 2026 forecast against realized numbers and quantify forecast accuracy. You may also propose your own direction with instructor approval.

Spec Category	Spec Details
Repository	A public GitHub repo containing all code, data, and writeup. Include a README that summarizes the project, lists dependencies, and explains the repo structure. Code should be reproducible from a clean environment.
Data	Document your data sources clearly. If you build a new dataset, include a data dictionary describing each variable, its type, and its role in your analysis. Note any preprocessing steps and justify your choices.
EDA	Produce visualizations that show the structure of your time series, including a monthly trend plot, an STL decomposition or seasonality check, and ACF/PACF plots. Confirm or challenge the previous team's findings about stationarity and seasonality. Two to four well-chosen plots beat ten cluttered ones.
Modeling	Fit at least two models: a baseline (seasonal naive is the convention) and your extended model. Justify your model choice with reference to your EDA. If you use auto.arima, explain whether you constrained it to seasonal models and why. Report AIC, BIC, and residual diagnostics (Ljung-Box at minimum).
Evaluation	Use a held-out test set with at least 8 months of data. Report RMSE, MAE, and MAPE. Compare your model to the seasonal naive baseline. State your quantifiable goal up front and report whether you achieved it.
Forecast & Interpretation	Produce a forecast for at least 12 months beyond your training data. Include 80% and 95% prediction intervals. Interpret what the forecast suggests about the labor market and acknowledge what your model cannot tell you.
Limitations	Devote a section of your report to honest discussion of your model's weaknesses. Public reporting bias, structural breaks, short test windows, and unmodeled covariates are all fair game. Identify at least three limitations.
Written Report	A PDF report of approximately 5 to 8 pages including figures. Sections should include: Goal Statement, Research Question, Data, EDA, Methodology, Results, Limitations, and Next Steps. Use first person. Write in plain language. Assume your reader is a smart non-specialist.
References	Cite all data sources and at least three additional sources (news articles, papers, or technical documentation) in IEEE format.

Deliverable Summary:

- GitHub repository (link submitted to Canvas)
- Written report (PDF, 5–8 pages)
- All code, data, and figures included in the repo